
EDUCATION

Program	Institution	%/CGPA	Duration
Ph.D., Computing Science	University of Alberta, Edmonton	-.-/4.0	Sep '25-Dec '29
B.Tech(DD), Electrical Engineering	Indian Institute of Technology Madras, Chennai	8.17/10.0	Nov '20 - May '25

PUBLICATIONS

- **ACTIVE FOUNDATIONAL MODELS FOR FAULT DIAGNOSIS OF ELECTRICAL MOTORS**
(Sriram Anbalagan, **Sai Shashank GP**, Deepesh Agarwal, Balasubramanian Natarajan, Babji Srinivasan)
 - Submitted to “[Engineering Applications for Artificial Intelligence](#)” Journal(Under Review)

SKILLS

- **Modelling & Analysis:** Autodesk Fusion 360, Autodesk EAGLE, MATLAB
- **Programming languages & Tools:** C, C++ , Python, MySQL, HTML, $\text{\LaTeX}$, Arduino IDE
- **Machine Learning:** Pytorch, Tensorflow, OpenAI-Gym, Scikit-Learn, Keras, OpenCV
- **Data analysis:** Pandas, Numpy, Matplotlib, Seaborn, Scipy

PROFESSIONAL EXPERIENCE

- **STUDENT RESEARCHER - [GOOGLE DEEPMIND](#)** SEP'24 - PRESENT
(Guide: [Dr. Aravindan Raghuv eer](#), Principal Research Scientist, Google DeepMind)
 - Formulating a novel algorithm to improve sample efficiency of RL algorithms through efficient reward design.
 - Using LLMs as a guiding tool to fabricate and modify the reward function based on Goal given in Natural Language.
 - Experimenting with SOTA LLM-enhanced RL to find out the weak spots to exploit and to improve the performance.
- **RESEARCH INTERN - [TCS RESEARCH](#)** MAY'24 - JUN'24
(Guide: [Dr. Prasant Misra](#), Senior Research Scientist, TCS Research)
 - Worked on joint optimisation of HVAC-EVSE energy consumption using offline reinforcement learning.
 - Collected custom datasets for the application using environments like SustainGym.
 - Developed near-optimal algorithm to using TD3 + behaviour cloning techniques using custom collected datasets.
- **ENGINEERING TRAINEE - [HIXAA TECHNOLOGIES PVT. LTD.](#)** DEC'21 - JAN'22
(Guide: [Atul Ghumade](#), Co-Founder of HIXAA Technologies Pvt. Ltd.)
 - Developed a guidance system for truck drivers to perfectly to align the trucks to silos' opening for easier unloading.
 - Enabled real-time positioning of container's opening using ArUco markers & blob detection in OpenCV.
 - Designed a UI which shows real-time positioning of the truck using Tkinter & assisted on paid user identification.

RESEARCH EXPERIENCE

- **YOUNG RESEARCH FELLOWSHIP PROGRAM - [YRF, IIT MADRAS](#)** AUG'22 - AUG'23
(Guide: [Dr. Babji Srinivasan](#), Department of Applied Mechanics, IIT Madras)
 - Worked on developing a deep learning framework to diagnose mechanical faults in Electric Motors.
 - Programmed the entire prediction framework using [TensorFlow](#) and ran experiments with real-world datasets.
 - Designed a novel method of active sampling which uses Shannon Entropy and KL-Divergence metrics.
 - Achieved an accuracy of **greater than 95%** across various datasets by using just 25% of labelled data.
- **OPTION INDEXING IN CONTINUAL LEARNING USING SUCCESSOR REPRESENTATIONS** JUN'23 - AUG'23
(Guide: [Dr. Balaraman Ravindran](#), Department of Computer Science, IIT Madras)
 - Extending the work done on Option Indexing([OI-HRL](#)) by Kushal Chauhan *et.al*.
 - Modified the given algorithm in the paper using Successor Representations.
 - Running experiments with the newly modified framework on [AI2THOR 3D](#) environments.
- **EXTENDING TARGET OPTIMISM TO CONTEXTUAL BANDITS WITH FAIRNESS GUARANTEES** SEP'23 - PRESENT
(Guide: [Dr. Balaraman Ravindran](#), Department of Computer Science, IIT Madras)
 - Innovating a new algorithm for contextual bandit setting which uses [Target Optimism](#) term introduced by Andrei *et.al*.
 - Formulating the new algorithm such that it satisfies the group fairness criteria formulated by Huang *et.al*. ([Link](#)).
 - Aiming to experiment the algorithm on real-world Hiring Processes to test its fairness.

TEACHING EXPERIENCE

- **TEACHING ASSISTANT FOR EE2703: APPLIED PROGRAMMING LAB** JUL'23 - NOV'23
(Course Instructor: [Dr. Nitin Chandrachoodan](#), Department of Electrical Engineering, IIT Madras)
 - Mentored a class of 180+ students on the concepts of Python Programming and its applications.
- **TEACHING ASSISTANT FOR CS6700: REINFORCEMENT LEARNING** JAN'24 - MAY'24
(Course Instructor: [Dr. Balaraman Ravindran](#), Department of Data Science and Artificial Intelligence, IIT Madras)
 - Mentored a class of 180+ students on the concepts of Reinforcement Learning and its theory.
- **TEACHING ASSISTANT FOR EE4371: DATA STRUCTURES AND ALGORITHMS** JUL'24 - PRESENT
(Course Instructor: [Dr. TG Venkatesh](#), Department of Electrical Engineering, IIT Madras)
 - Mentoring a class of 60+ students on theory and implementations of Data Structures and Algorithms handling them.

COURSE PROJECTS & PAPER PRESENTATIONS

- **SOLVING TAXI-V3 ENVIRONMENT USING HIERARCHICAL REINFORCEMENT LEARNING (HRL)** APR'23
(Course Instructor: [Dr. Balaraman Ravindran](#), Department of Computer Science, IIT Madras)
 - Utilized Semi-MDP Q-Learning and Intra-Option Q-Learning frameworks of HRL to solve the environment.
 - Programmed the entire framework from scratch using Numpy and compared the different approaches.
- **THOMPSON SAMPLING IN CASCADED BANDITS - PAPER PRESENTATION** NOV'23
(Course Instructor: [Dr. Kota Srinivasa Reddy](#), Department of Electrical Engineering, IIT Madras)
 - Reviewed the paper "A Thompson Sampling Algorithm for Cascading Bandits" by *Cheung et.al.* ([Link](#))
 - Explained the mathematical proof and recreated the results mentioned in the paper.
- **GAUSSIAN PROCESS OPTIMIZATION IN BANDIT SETTING - PAPER PRESENTATION** APR'23
(Course Instructor: [Dr. Andrew Thangaraj](#), Department of Electrical Engineering, IIT Madras)
 - Reviewed the paper "Information-Theoretic Regret Bounds for Gaussian Process Optimization in the Bandit Setting" by *Niranjana et.al.* ([Link](#))
 - Explained the mathematical proof using the concepts taught in the course "Information Theory".
- **ELECTRIC CIRCUIT SOLVER USING PYTHON** FEB'22
(Course Instructor: [Dr. Harishankar Ramachandran](#), Department of Electrical Engineering, IIT Madras)
 - Programmed an API to solve for all the unknowns in a basic analog circuit with both DC & AC components.
 - Created the API using classes & Numpy's matrix functionalities.

PERSONAL PROJECTS

- **HOGWARTS CHESSBOARD** JUL'21 - NOV'22
(Part of [Electronics Club](#), IIT Madras)
 - Transformed generic chessboard with movement automation & remote connectivity for enhanced game experience.
 - Devised cloud-based communication system via RPi4 to exchange players' responses & move pieces in reality.
 - Implemented stepper motor system to realise 2D plotter mechanism for movement of magnet embedded chess pieces.

RELEVANT COURSES

- | | |
|--|--|
| ● Reinforcement Learning | ● Linear Algebra for Engineers |
| ● Fundamentals of Deep Learning | ● Probability for Electrical Engineers |
| ● Multi-Armed Bandits | ● Graph Theory |
| ● Pattern Recognition & Machine Learning | ● Applied Programming Lab |
| ● Information Theory | ● Computer Methods in EE |
| ● Convex Optimisation | |

ACHIEVEMENTS AND AWARDS

- Achieved a **percentile of 99.91** in the IIT-JEE Main 2020 among 1.4 million applicants
- Achieved a **percentile of 99.6** in the IIT-JEE Advanced 2020 among 200 thousand candidates
- Received Kishore Vaigyanik Pratishthan Yojana Fellowship 2019 from the Government of India
- Received Young Research Fellowship 2022 from Indian Institute of Technology Madras
- Attended a fully sponsored AI Alignment Workshop conducted by MIRI, UC Berkeley

POSITIONS OF RESPONSIBILITY

- EVENTS HEAD - [SHAASTRA, IIT MADRAS](#)

JUN'22 - FEB'23

- Spearheaded a **team of 15 members** in organising 5 tech competitions in Shaastra 2023.
- Played an integral role in negotiating with companies like [Bajaj Auto](#) to sponsor our circuit design competition.

- EVENTS COORDINATOR - [SHAASTRA, IIT MADRAS](#)

AUG'21 - MAR'22

- Ideated and organised events based on PCB Designing & all-round electronics for Shaastra 2022
- Managed a footfall of **500+ participants** hailing from all over the country.

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- WORKSHOP TRAINER - SAMPARK, IIT MADRAS

- Conducted a hands-on introduction to machine learning workshop
- Marketed our Shaastra workshop to an audience of 75+ students

- WORKSHOP TRAINER - [SHAASTRA, IIT MADRAS](#)

- Taught building supervised learning models to an audience of 100+
- Generated a revenue of 50k INR from the workshop

- STUDENT MENTOR - [SHIKSHA PRAYAS NGO](#)

- Coached two 9th standard students for their all round development
- Helped them improve their academic score than they initially expected